Al-Balqa Applied University
ComputerEngineering Department
Student Name:

Assembly Language Programming
Midterm Exam (40%)

Faculty of Engineering Technology
Sunday 11/11/2018
Instructor:

Q1) Fill the table with the correct answer(12 marks)

1	2	3	4	5	6	7	8	9	10	11	12

1) Given that register CH contains an unsigned number N. What are the minimum and maximum possible values of N, respectively?

A) 0 and 127. B) 256 and 255.C) -128 and 128 D) 0 and 255

2) If 1110 0100 is the 8-bit 2's complement representation of a number N, what is the decimal value of N?

A) -28 B) +228 C) -228 D) -74

3) Which of the following summation set the CF and SF?

A) 7FF1h
+000Fh

B) 7FFEh
+ 0001h

C) FF00h
+ FF00h

D) 8000h
+ 8000h
---

4) Which of the following statement is correct, if X DB 22H

A)MOV [SP],99H B) MOV BL,OFFSET X C)PUSH XD)none of them

5) What is the type of addressing for the following statement: MOV BX,[SI]

A) Indexed B) indexed based C) direct D) based

6)If you want to compare the following two signed values AX = 123Fh, BX = B3F0h and put the value 1 in CL if AX < BX, you write:

A)
```
CMP AX, BX
JL L1
.....
L1: MOV CL, 1
```

B)
```
CMP AX, BX
JA L1
.....
L1: MOV CL, 1
```

C)
```
CMP AX, BX
JB L1
.....
L1: MOV CL, 1
```

D)
```
CMP AX, BX
JG L1
.....
L1: MOV CL, 1
```

7)The value of CX after executing the following code is:

```
        MOV AL, 0FFH
MOV CX, 5
L1: ADD AL, 1
        LOOPZ L1
```

A) CX=0 B) CX=5 C) CX= 4 D)CX=3

Zerone Team Computer & Networks Engineers

Scanned by CamScanner

8) What is the value of the AL register after executing the following code?

```
MOV AX, 0FF02H
MOV CL, 0FCH
XCHG AL, CL
ADD AL, 5
SBB AL, 1
```

A) 01H B) 00H C) FFH D) None of them

9) Number of the times the instruction sequence below will loop before coming out of loop is

```
MOV AL, 00H
L1:INC AL
JNZ L1
```

A) 0 B) 1 C) 256 D) Infinite Loop

Assume: SS=1000H,DS=7000H,SI=0020H,DI=0100H,X=0500H,BP=0300h

10)What is the physical address of value that will be stored in DH for the following instruction MOV DX,X[SI+2]

A) 7522h B) 10522h C) 70522hD)70523h

11)Calculate the physical address of the destination operandfor the following instructionMOV [BP + 40H], CL

A)70340h B) 10340h C)1340h D)7340h

12)What is the value of the SP register after executing the following instruction, if the SP =FFFAH?POP AX
DEC SP
PUSH BX

A) FFF7H B) FFF8H C) FFFBH D) FFF9H

Team
ZeroOne
Computer & Networks Engineers

Q2) Trace the following codes and find the result of executing by filling the blanks. (25 marks)

1)
```
MOV CX,8
MOV BL,2
MOV AL,9
L:MOV AH,0
DIV BL
CMP AL,0
LOOPNZ L
```

AH= AL= BL= CX=

Scanned by CamScanner

2)MOV AX,008AH
MOV DX, F380H
MOV BL,2
ADD AL,DL
CMP DH,2
JL L
DIV BL
JMP L1
L: MUL BL
L1: HLT

AH= AL= DH= DL=

3) X DW733H,1122H
MOV AX,4455H
PUSH AX
PUSH X+1
POP AX
POP BX

AH= AL= BH= BL=

4) X DB 3,5,7
INC X+1
MOV AX, WORD PTR X+1
NEG AX

AL= AH= CF= SF= PF= AF=

Team ZeroOne Computer & Networks Engineers

5) X DW 5437H
Y DW 28H
MOV BX, OFFSET X
ADD BX,1
MOV AL,[BX]
MOV CL,[BX+1]
ADD AL,CL
INC BX
MOV DL,[BX]

DL= AH= AL= CL=

Scanned by CamScanner

6) X DB 3,-5,4
MOV CX,2
MOV SI,1
L2:TESTX[SI],81H
JZ L1
NOT [SI]
L1:INC SI
LOOP L2

X= ---------,---------,---------

Q3)

a) Suppose we have a microprocessor with 24 bits address bus, what is the maximum memory capacity will be?(1 mark)

b)Write the equivalent assembly codes to the following C++ codes: (3 marks)

Note: x, y and s are unsigned byte variable.

```
if( x > y)
 s=s+x;
else
 s=s-y;
```

Team
Zerone
Computer & Networks Engineers

Scanned by CamScanner

Q#1:[12]Circle the correct answer:

1. Data bus width of Intel 8086 microprocessor is

(a) 16 (b) 24 (c) 20 (d) 32

2. The length of extended ASCII codes

(a) 7 bits (b) 8 bits (c) 12 bits (d) 16bits

3. One of the following is not part of execution unit of 8086 microprocessor

(a) BP register (b) AX register (c) DS register (d) SP register

4. The equivalent code of NEG AX instruction is:

a) XOR AX,AX
SUB AX,1

b) SUB AX,AX
ADD AX,1

c) OR AX,AX
ADD AX,1

d) XOR AX,FFFFH
ADD AX,1

5. A 16-bit address bus allows access to a memory of capacity

(a) 128KB (b) 64KB (c) 256KB (d) 512KB

6. If BX=8321h, CF=1 and cl=5 after execution ROR BX,CL the:

a) BX= 0C19H and CF=1 b) BX=0C19H and CF=0

c) BX=1832H and CF=0 d) BX=1832H and CF=1

7. One of the following control transfer instruction is not used with signed number

a) JGE b) JBE c) JL d) JLE

8. . After execution of the following code register AL contains

```
MOV  AL, 'a'
ADD  AL , 2
AND  AL, 1101 1111b
```

a) AL='c' b) AL='C' c) AL='a' d) AL='A'

ZeroOne Team
Computer & Networks Engineers

Scanned by CamScanner

10. If the SP register contains the decimal value 340, what will be its value after executing the following program segment?

```
PUSH AX        MOV BX, 25
PUSH BX        MUL BX
PUSH DX        POP BX
```

a) 344 b) 338 c) 342 d) 336

11. Given that register CH contains an unsigned number N. What are the minimum and maximum possible values of N, respectively?

a) 0 and 127.
b) 256 and 255
c) −128 and 128.
d) 0 and 255

12. If 1110 0100 is the 8-bit 2's complement representation of a number N, what is the decimal value of N?

a) −28.
b) +228.
c) −228.
d) −74.

ZerOne Team
Computer & Networks Engineers

Q2: [15 points] Trace the following programs and determine the final value for each register:

a)
```
MOV AL,05H
NEG AL
MOV BL,02H
AND BL,AL
```

BL=	AL=	PF
2	FB	0

a)
```
MOV BL,07H
SUB BL,0EH
MOV DL,02H
MOV AL,FFH
ADD AL,1
JC L
MOV CL,FFH
JNC LL
L: MOV CL,00
LL: ADC DL,1
```

AL=	BL=	CL=	DL=
00	F9	00	02

c)
```
MOV CL,03H
MOV AX,127
ROR AX,01H
JC L
ADC CL,02H
L: HLT
```

AX=	CL=	CF=	ZF=	PF=
803F	03	1	0	1

Scanned by CamScanner

a) MOV AX,35H
MOV BX,39H
PUSH AX
PUSH BX
ADD AX,BX
DAA
MOV DX,AX
POP AX
POP BX

AX=	BX=	DX=
35	39	34

b) MOV AL,14
MOV AH,0
MOV BL,17
DIV BL

AL= 0.82 AH= 00

Team
ZeroOne
Computer & Networks Engineers

uestion #2

uppose we have a microprocessor with 24 bits address bus:

a) What is the maximum memory capacity will be?

ximum mory = 2^{24} 2

b) For the following instruction MOV [BX+76Dh],AL

1- What is the addressing mode used ?

Base addressing mode with displacement 1.5

2- Calculate the physical address of memory location assuming that DS=8CFBh and BX= 4000h

ph address = rtg x10h + off [5 points]

phy add = 8CFBh x10h + 4000h

= 90FB0
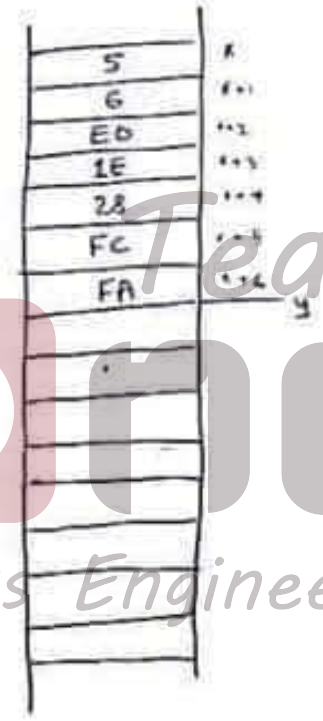
Scanned by CamScanner

Question #3

Write an assembly code to clear low order 4 bit of memory locations and store the result into Y. [4 points]

```
X    DB   5,6,-20,30,40,1,-4,-6
Y    DW   8 DUP(0)

     Mov AH,[X]
     Loop
     Mov Y,AH
     HLT
```



Question #4

Given 256 KB of memory locations , Write an assembly code to find the sum of memory locations that h values greater than 5. [4 points]

ZerOne Team
Computer & Networks Engineers

Scanned by CamScanner

Question #4

Choose the correct answer:

1. Address bus width of Intel 8086 microprocessor is

(a) 16 (b) 24 (c) 20 (d) 32

2. The length of ASCII codes

(a) 7 bits (b) 8 bits c) 12 bits (d) 16bits

3. One of the following is a part of bus interface unit of 8086 microprocessor

(a) BP register (b) AX register (c) DS register (d) SP register

4. The equivalent code of NOT AX instruction is:

a) XOR AX,FF
ADD AX,1

b) SUB AX,AX
ADD AX,1

c) XOR AX,AX
ADD AX,1

d) XOR AX,FFFFH

5. A 34-bit address bus allows access to a memory of capacity

(a) 16MB (b) 64MB (c) 4GB (d) 16GB

6. One of the following control transfer instruction is not used with unsigned number

(b) JGE b) JBE c) JA d) JAE

7. The stack pointer always points to a(n) ... location.

a) full b) empty c) byte d) None of the above

8. If the SP register contains the decimal value 340, what will be its value after executing the following program segment?

```
PUSH AX        MOV BX, 25
PUSH BX        MUL BX
PUSH BX        POP BX
```

a) 344 b) 338 c) 342 d) 336

9. Given that register CH contains anunsigned number N. What are the minimum and maximum possible values of N, respectively?

a) 0 and 127 b)0 and 128 c) -128 and +127 d) 0 and 255

10. If 11111110 is the 8-bit 2's complement representation of a number N, what is the decimal value of N

a) -2 b) 254. c) -3. d) 1.

ZeroOne Team
Computer & Networks Engineers

Scanned by CamScanner

QUESTION #3

a) Write an equivalent Assembly code for the following statement

If X> 0 res = ((X*10)+ Y)-Z else res= X-2

Note: X, Y and Z are byte sizes, value unsigned; res is word size.

```
XDB 2
YDB H
ZDB 6
res DW ?
mov cl,10
mov AL,X
cmp AL,0
JBE L1
mul cl
```

```
ADD AX,Y
SUB AX,Z
JMP L2
L1: SUB AX,2
mov res,AX
L2: HLT
```

b) Write an equivalent of the following instructions:

```
LOOP L
```

```
L1: mov AL,H
dec cx
cmp cx,0
JNE L1
```

```
NOT BL
Neg BL
Dec BL

SUB AL,BL
Neg BL
ADD AL,BL
```

c) Write instructions to test bit-2 in register AL. If it is one reset it otherwise invert all bits in register AL.

```
AND AL,04
JNE L1
Not AL
JumL2
L1: AND AX,FD
L2: HLT
```

d) Write an assembler codes to push into the stack even numbers from 0-100

```
mov cx,100
mov Si,0
L2: AND X[Si],01
JNZ L1
push X[Si]
INC Si
Dec cx
JNZ L2
JumL3
L1: Jmp L2
```

```
L2: AND X[Si],01
```


Team ZeroOne Computer & Networks Engineers

Scanned by CamScanner

Al-Balqa Applied University Faculty of Engineering Technology Computer Engineering Department	*Microprocessors and Assembly Language MID Exam*	Date: Instructor Name: Student Name:

QUESTION #1

Find the output of the following program segments:

a)

```
X DB 1H,4H,-3H
Z DW 1030H

MOV AL,[X+2] ;AL = FD
MOV AH,30h ; AH = 30
AND Z,AX ;Z = 1030 H
NEG Z
HLT
```

AX= 30FD ; Z= EFD0

b)

```
X DD BACF0123H,0F5432DH

MOV SI,0
MOV BX, OFFSET X
MOV AL,[BX] ; AL= 23
INC BX
INC BX
MOV AH,[BX] ; AH= CF
MOV DX,[BX+3] ; DX = F5 43
XCHG DL,X[SI] ; DL= 23 43
```

AX= CF23 ; DX= F543 X= 43 H

c)

```
MOV CL,04H ; CL=04H
MOV AL,-7 ; AL= F9
ROR AL,CL ; AL= 9F
JNC L
ADC CL,01H ; CL= 06H
JMP L2
L:MOV DL,AL
AND AL,0FH
L2:HLT
```

AL= 9F H ; DL=06H ; ZF= 0

d)

```
X DW 501FH
   MOV AX,35H ; AX= 35
   MOV CL,8 ; CL=08H
   MOV BX,39H ; BX=39H
   PUSH AX
   PUSH X
   INC X ; 5020
   POP BX ; BX= 501F
SHR AX,CL ; AX=0000
   POP DX ; DX = 0035
   AND DX, BX ; DX= 0015
```

AX= 0000 H ; BX= 501F ; DX= 0015 X= 5020 H

QUESTION #2

Assume that you have a microprocessor called ASSEM_MICRO, the address bus size equal 14 bits the data bus size 8 bits and the register size is 12 bits: CS=600H; IP=50H :
Calculate the physical address for the instruction pointed by IP

Ph = 600 * 16 + 50 = 2050

Team ZeroOne Computer & Networks Engineers

Scanned by CamScanner

Faculty of Engineering Technology

Microprocessors and Assembly Language (Mid Exam)

STUDENT NAME: ..... Marks(40)

Q1: Multiple Choice Questions: (18 pts)

1- The 16 bit flag reg of 8086 microprocessor is responsible to indicate:
A. the condition of result of ALU operation B. the condition of memory C. the result of subtraction

2- The CF is known as:
A. condition flag B. carry flag C. common flag

3. The SF is called as _____
A. service flag B. sign flag C. single flag

4. The IP is _____ bits in length
A. 8 bits B. 4 bits C. 16 bits

5. To exchange 2 bytes stored at CL and DL regs:
A. xchange DL,CL B. xchg DL,CL C. MOV DL,CL

6. The result of MOV AL, 65 is to store
A. store 0100 0010 in AL B. store 42H in AL C. store 0100 0001 in AL

7. Which of the following instruction is not valid?
A. MOV AX, BX B. MOV DS, 5000H C. MOV SI, 5000H

8. In PUSH instruction, after each execution of the instruction, the stack pointer
A. incremented by 1 B. decremented by 1 C. incremented by 2

9. Which of the following instruction is correct?
A. MOV X, AX B. LEA BL, X C. MOV DX, [AX]

10. Imul instruction treats the FCH value as :
A. 4D B. -4D C. 11111110B

Team Zerone Computer & Networks Engineers

Scanned by CamScanner

Q3) Write the type of addressing mode of the following code example (5PTS.)

a. MOV BL,24H. → Immidand addresing mode

b. MOV AL, [3000H] → direct addresing mode

c. MOV BX, [SI] → index addresing mode

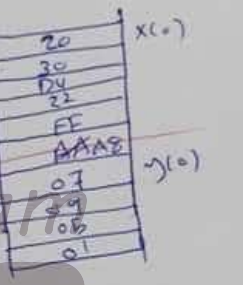
d. MOV CX,[BX+DI] → Base index addresing mode

e. MOV CX, ARRAY[DI] → Indirecit addresing mod

Q4) Trace the following: (10 PTS.)

1)

```
X DW 3020H,22D4H,0AAFFH
Y DB 7,9,5,1
MOV BX, OFFSET X
MOV AX,[BX]
ADD BX,2
MOV CX,[BX]
MOV SI, OFFSET Y
SUB SI,1
SUB [SI],2
```



AX= 3020H CX= 22D4H DX= X= 3020H

2) MOV BL,-122

ADD BL, 7

BL= 8D H SF= 1 PF= 1 OF= 0

3) MOV AL, 18

MOV BL, 10

MUL BL

DEC BL

AX = ...B.H. BL= ....9.... H

Team
Cerone
Computer & Networks Engineers

Scanned by CamScanner

11. What is the possible decimal value could the 8-bit number 9Eh represent?

A) 158 B) -158 C) -98

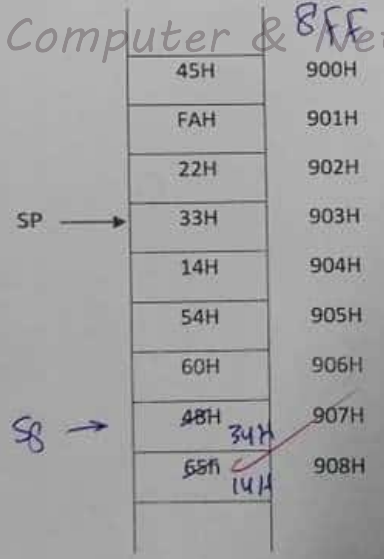
12. The effect of the following instructions on the ax register is:

```
push ax
add ax, 4
pop bx
mov cx, ax
push bx
pop ax
```

A. leave it with its original value B. add 4 to it C. clear it

Q2) Write suitable instructions to do the following (7 pts.)

A. If sp= 903h , INCREMENT the word pointed by sp in the stack, then store it back in stack location address 907h



B. After executing branch a, what is the stack after applying this code :

Push 1430H

Push 6589H

ZeroOne Team
Computer & Networks Engineers

Scanned by CamScanner

b) Complete the following table: (4)

Instruction	Result	CF	ZF	SF	OF	PF	AF
MOV BH, 125	BH = 7DH	0	0	0	0	0	0
ADC BH, 6	BH = 83H	0	0	1	0	0	1
SBB BH, BH	BH = 00H	0	1	0	0	1	0
NEG BH	BH = 00H	0	1	0	0	1	0

Third Question (12 marks):

a) (5 marks) For the following stack memory, fill-in the blanks with the correct answer:

1) SUB SP, 4
PUSH 205Bh

SP = 014C H

Show the results in the stack blocks

SP →	5B 20	0500:014Ch
	20 10	0500:014Dh
	CD 55	0500:014Eh
	FE 01	0500:014Fh
	FF	0500:0150h
	49	0500:0151h
SP →	D0	0500:0152h
	39	0500:0153h

2) POP DX
POP AX

DX = 205B H.

AX = FECD H

3) AND AX, 5 → AX = 0005H
PUSH AX

AX = 0005 H.

SP = 014E H

Show the results in the stack blocks

(5)

Team ZeroOne
Computer & Networks Engineers

b) (7 marks) Consider the following initial register values: DS=C00h, ES=3000h, SS=600h, SP=1009h, and the initial memory locations values are as shown in the figure. Answer the questions below:

1) For the instruction MOV SI, [BP-1], calculate the physical address needed to read the data from memory - if any:

Logical Address = 0600 : 100A

Physical Address Calculation: 700A

physical add = 0600 * 10h + 100A

= 700A h

SS →

7009h	24
700Ah	DE
700Bh	CC
700Ch	09
C0DEh	22
C0DFh	AE
C0E0h	11
C0E1h	80
C0E2h	07

DS →

ES →

Scanned by CamScanner

29.5 31.5

40

Al-Balqa Applied University		Date:
Faculty of Engineering Technology		Instructor Name: [redacted]
Computer Engineering Department		Student Name: [redacted]

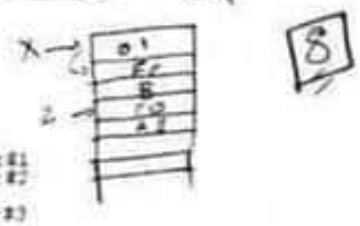
First Question (10 marks): Fill-in the table with the correct answer:

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
B	A	A	D	C	X	A	D	A	B

Consider the following code: Answer questions 1 - 3

```
X DB 1, -1, 8
Z DW 0ABCDh

MOV AL, X-1
AND byte PTR Z, AL
NEG Z
```



Q1) The value of AL after executing (inst#1) is:

a) 1 b) FCh c) 8 d) Error

Q2) The value of memory location Z after executing (inst#2) is:

a) 0CCh b) 0ABCDh c) 0ABCCh d) 0CCABh

Q3) The value of Z after executing (inst#3) is:

a) 5434h b) 5433h c) 3454h d) 3354h

Q4) What is the value of registers AX and BX after executing the following code are:

```
MOV AL, 10H
MOV BX, 1204H

XCHG BX, BX
```

a) AX = 10H, BX = 1204h b) AX = 01A4h, BX = 1208 c) AX = 1204h, BX = 010B d) AX = 1204h, BX = 0002h

Consider the following data segment memory locations, answer questions from 5-8

0700:0102h	10 ← X
0700:0103h	FF
0700:0104h	48
0700:0105h	23
0700:0106h	1F ← Y
0700:0107h	78
0700:0108h	67
0700:0109h	CD

Q5) The value of BX after executing the following instruction is:

MOV BX, offset Y

a) 1Fh b) 0700:0106 c) 0106h d) 0700h

Q7) The value of DH after executing the following instruction is:

MOV DH, [Y-2]

a) 48h b) 0104 c) 0700:0104 d) Error

ZeroOne Team Computer & Networks Engineers

Scanned by CamScanner

2) The value of DI after executing the instruction in 1 is:

DI : DEH

3) The value of DL after executing the following instruction is:

MOV DL, [DI+1]

DL : AEH

4) Give an instruction on each of the following addressing modes:

1- Direct addressing mode: MOV AL, 3

2- Based with displacement addressing mode: MOV AH, [BP-1]

3- Based indexed addressing mode: MOV BX, [BP+SI]

Forth Question (4 marks): Write an equivalent Assembly code for the following equation:

res = ((X-10) * Y)/Z

Note: X, Y and Z are byte sizes

```
X db ?
Y db 5
Z db 2
res db ?

MOV AL, X
MOV BL, Y
MOV CL, Z
add AL, 10
MUL BL
div CL
MOV res, AX
HLT
```

3.5

ZeroOne Team
Computer & Networks Engineers

Scanned by CamScanner

Q8) The value of X after executing the following instruction is:

```
LEA DI, X
MOV [DI], 15
```

a) 10h b) 15 c) 15h d) 0Fh

Q9) Consider the following instructions:

```
arr DB 3 dup(?)
MOV arr[0], 12h
MOV arr[1], -2
MOV arr[2], F9h
```

The values of arr after executing the code are:

a) 12, FE, F9 b) 12, -2, F9 c) 12, 0FC, 0F9 d) Error

Q10) The instruction that is equivalent to the following operation is:

```
MOV AX, 0
```

a) AND AX, AX b) XOR AX, AX c) OR AX, AX d) NOT AX

Second Question (14 marks):

a) Trace the following codes and fill-in the spaces:

1)

```
MOV AL, 253
CMP AL, 0
JL L1
MOV CL, 0
JMP L2
L1: MOV CL, 1
L2:
```

CL = _____ H

2)

```
MOV AL, -9
MOV BL, 2
IDIV BL
```

AX = _____ H

3)

```
MOV AL, 128
MOV BL, 8
MUL BL
```

AX = _____ H

4) Consider that CF = 1

```
MOV AX, 1
MOV BX, 1F3Dh
SBB AX, BX
```

AX = _____ H

5) Consider that: 'a' = 61h, 'A' = 41h

```
ARR DB 'abAaD'
MOV SI, 0
MOV CL, 5
L2: CMP ARR[SI], 61h
JNE L1
SUB ARR[SI], 20H
L1: INC SI
DEC CL
CMP CL, 0
JA L2
```

ARR = _____

6) Consider that the logical address for X is 0700:0114h

```
X DB 12, -3, 8, 5
Y DB 0

LEA BX, X
L2: TEST [BX], 80h
JZ L1
INC Y
L1: INC BX
CMP BX, OFFSET X+4
JNZ L2
```

BX = _____ H

Y = _____ H

ZeroOne Team
Computer & Networks Engineers
Scanned by CamScanner